

NETWORKS AND COMPLEXITY

Exercise Sheet 12: Models of Growth

*This is an exercise sheet from the forthcoming book Networks and Complexity.
Find more exercises and solutions at <https://github.com/NC-Book/NCB>*

Ex 12.1: Integration [Lvl. 1]

Find the general solutions for the following differential equations:

a.

$$\dot{x} = 4x$$

b.

$$\dot{x} = \frac{1}{2x}$$

c.

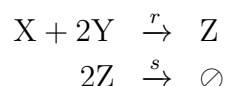
$$\dot{x} = \frac{x}{a+t}$$

d.

$$\frac{dc_k}{dk} = -\frac{2c_k}{k}$$

Ex 12.2: Mass action laws [Lvl. 1]

Write the differential equations for x , y and z that correspond to the reaction diagram



Ex 12.3: Infinite excess degree [Lvl. 1]

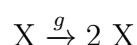
Consider a network with $p_0 = 0$ and $p_k = ak^{-3}$ for $k > 0$. Show that this network has finite mean degree but infinite mean excess degree. You can use the $\sum_{k=1}^{\infty} k^{-2}$ is finite while $\sum_{k=1}^{\infty} k^{-1}$ diverges. [Hint: Don't try to determine a .]

Ex 12.4: Population growth [Lvl. 2]

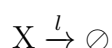
Sexual reproduction requires that two individuals of a species meet. If we apply the law of mass action naively we may arrive at a model of the form $\dot{x} = x^2$. Solve the initial value problem for $x(0) = 1$ and explore what happens.

Ex 12.5: Birth and death [Lvl. 3]

A population of bacteria X grows by cell division at rate g



In addition the bacteria die spontaneously at rate l



Derive an ODE for the population size x and solve it.

Ex 12.6: The leaky sink [Lvl. 3]

Water is flowing into a sink at a rate a liters per second. But the stopper that is supposed to plug the outflow at the bottom does not fit well such that a little bit of water leaks out. The rate of leakage is proportional to the water pressure at the bottom of the sink. Hence, the outflow is b liters per second for every liter of water that is currently in the sink.

- Write a differential equation for x , the volume of water that is in the sink.
- Find the steady state.
- Find the general solution to the differential equation. [Hint: If stuck, consider $\delta = x^* - x$.]
- Solve the initial value problem for a sink that is initially empty.

Ex 12.7: Condensation [Lvl. 3]

Consider a bathroom mirror on which droplets of condensation are forming. The drops grow by absorbing water vapor on their surface. Hence the rate at which a droplet absorbs volume is proportional to the radius r of the droplet squared. Because the droplet is a three-dimensional object its volume scales as the r^3 . Hence it is reasonable to model the growth of the droplet by

$$\dot{v} = ar^2 = bv^{2/3} \quad (81)$$

where a and b are parameters and v is the volume. Solve the initial value problem with $v(0) = 0$. Is there only one solution?

Ex 12.8: Iterative integration [Lvl. 3]

Consider the equation

$$\dot{x} = rx$$

where $x(0) = x_0 = 1$. We know that we can't solve this equation by direct integration. However, let's try nevertheless ...

- Assume that x on the right hand side of the equation is a constant and directly integrate the equation. (This will prove the assumption wrong)
- Now take your solution from (a) and use this as a new assumption, for x and integrate again.
- Iterate this process to find successively better approximations to the solution. Can you spot the pattern that is developing?

Ex 12.9: Price's model [Lvl. 3]

When a scientific paper is published it cites m other papers. Papers that have already been cited more often have higher visibility and hence attract new citations at a higher rate. We assume that a paper with k citations has the visibility $v = a + k$, where a represents an intrinsic visibility independent of citations.

- Let p_k be the in-degree distribution of the papers, the proportion of papers that have been cited k times. Consider a single citation from a new paper. Assume that the chance that a paper is cited is proportional to its visibility. Show that the probability v_k that a single new citation cites a paper that has previously had k citations is

$$v_k = p_k \frac{a + k}{a + z}$$

- Show that in the long run the in-degree distribution p_k approaches a power-law for $k \gg 1$, and determine the power-law exponent.

Ex 12.10: Change of variables [Lvl. 3]

Consider a system of differential equations where

$$\begin{aligned}\dot{x} &= y \\ \dot{y} &= -x\end{aligned}$$

Transform these equations into polar coordinates

$$\begin{aligned}r &= \sqrt{x^2 + y^2} \\ \phi &= \arctan(y/x).\end{aligned}$$

Then solve the differential equations.

Ex 12.11: Box of bolts [Lvl. 4]

In a factory there's a big box of bolts, some of which are broken. At a rate r a worker takes a bolt that is not broken out of the box. Derive a differential equation for the proportion of bolts that are broken, then solve it. Check your result by deriving the same solution in a different (perhaps simpler?) way.

Ex 12.12: Logistic growth [Lvl. 4]

Many systems in nature exhibit logistic growth, which is described by the logistic differential equation $\dot{x} = Ax(K - x)$.

- Show that the equation for the SIS model can be written in the form of logistic growth and determine the values of A and K .
- Show that the logistic differential equation can be solved by separation of variables, identify the two integrals that need to be solved, and solve the simpler one of them.
- We are now left with a more difficult integral to solve. To do this, first show that the term under the integral can be written in the form

$$\frac{a}{x} + \frac{b}{K - x}$$

where a and b are constants. In other words: [Hint: After the first steps, isolate the x .]

- Use the result from (c) to split the integral that we still need to solve into two integrals and solve them.
- Finally put all the parts together and solve for x .
- Substitute K and A into the solution to find a general solution for the SIS model.